

ALMOST SURE ASYMPTOTIC REPRESENTATIONS OF THE TRIMMED WILCOXON STATISTIC AND TRIMMED HODGES-LEHMANN ESTIMATOR WITH APPLICATIONS

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SUMMARY

In Section 1 the trimmed Wilcoxon one-sample statistic is defined and an asymptotic a.s. representation in terms of an average of i.i.d. random variables is derived (cf. Hettmansperger (1968) who defined a trimmed Mann-Whitney statistic and proved its asymptotic normality). Next a trimmed Hodges-Lehmann estimator is defined and also expressed in terms of an average of i.i.d. random variables using the above representation (cf. Salch (1976) who also defined a trimmed Hodges-Lehmann estimator, but only proved its asymptotic normality). Asymptotic relative efficiencies of the estimate are given for various distributions. As an application a fixed-width sequential confidence interval procedure for the unknown median of a symmetric distribution is defined and analyzed asymptotically in Section 2. Some Monte Carlo results are also given. As another application a sequential selection procedure is given in Section 3.

1. Almost sure asymptotic representations

Let $X_1, X_2, X_3, \dots$ be a sequence of independent random variables with common cdf. $F_\theta(x) = F(x - \theta)$, where F is symmetric about 0 and has density f . In the rest of this section it will be assumed without loss of generality that $\theta = 0$. Let

$$(1) \quad r = [\alpha n] + 1, s = [(1 - \alpha)n]; F(g_1) = \alpha, F(g_2) = 1 - \alpha \text{ for } 0 \leq \alpha < \frac{1}{2}$$

where $[x]$ is the largest integer less than or equal to x .

The following assumptions will be used throughout the present section:

(A) f is continuous in neighbourhoods of g_1 and g_2 ; $f(g_1) > 0$ and $f(g_2) > 0$

(B) $\int_{g_1}^{g_2} f^2(x) dx < \infty$

(C) F has a second derivative in some neighbourhoods of g_1 and g_2 with F'' bounded in these neighbourhoods.

- (D) $U(x)$ has a second derivative in some neighbourhoods of 0 with U'' bounded in the neighbourhood, where

$$U(x) = P \{ (X_1 + X_2)/2 \leq x, g_1 \leq X_1 \leq g_2, g_1 \leq X_2 \leq g_2 \}$$

For the proofs of Theorems 1, 2 and 3 below, we first consider the following lemmas:

Lemma 1

Assumptions (A) and (B) guarantee the existence of $U'(x)$ in some neighbourhood of 0. Furthermore, if $f(x)$ has a Radon-Nikodym derivative $f'(x)$ for $x \in [g_1, g_2]$ satisfying

$$(2) \quad \int_{g_1}^{g_2} |f'(x)| dx < \infty \text{ and } \int_{g_1}^{g_2} (f'(x))^2 dx < \infty,$$

then (D) follows from (A).

For brevity, the proof of the lemma is omitted.

Let $X_{(1)} \leq \dots \leq X_{(n)}$ be the order statistics of $X_1, X_2, \dots, X_n$ and define

$$(3) \quad F_n(x) = n^{-1} \sum_{k=1}^n I(X_k \leq x).$$

Let $\{a_n: n = 1, 2, \dots\}$ be a sequence of positive constants such that

$$(4) \quad a_n \sim 2(\log n)/n^{1/2} \text{ as } n \rightarrow \infty$$

Lemma 2

Under (A) and (C)

$$(i) \quad H_n^{(1)} = O(n^{-1/2} \log n) \text{ a.s. as } n \rightarrow \infty$$

$$(ii) \quad H_n^{(2)} = O(n^{-1/2} \log n) \text{ a.s. as } n \rightarrow \infty$$

$$(iii) \quad X_{(r)} = g_1 + O(n^{-1/2} (\log 2n)^{1/2}) \text{ a.s. as } n \rightarrow \infty$$

$$(iv) \quad X_{(s)} = g_2 + O(n^{-1/2} (\log 2n)^{1/2}) \text{ a.s. as } n \rightarrow \infty$$

where

$$H_n^{(1)} = \sup_{x \in I_n^{(1)}} |G_n^{(1)}(x)|, \quad I_n^{(1)} = (g_1 - a_n, g_1 + a_n)$$

$$G_n^{(1)}(x) = [F_n(x) - F_n(g_1)] - [F(x) - F(g_1)]$$

$$H_n^{(2)} = \sup_{x \in I_n^{(2)}} |G_n^{(2)}(x)|, I_n^{(2)} = (g_2 - a_n, g_2 + a_n)$$

$$G_n^{(2)}(x) = [F_n(x) - F_n(g_2)] - [F(x) - F(g_2)]$$

Proof

The proof of (i) and (ii) follows from Lemma 1 of Bahadur (1966), while (iii) and (iv) result from Lemma 3 of Bahadur (1966) and the law of the iterated logarithm.

Lemma 3

Under (A) and (C)

$$(i) \quad \sum_{i=1}^n I(-X_{(r)} - a_n \leq X_i \leq g_2) = O(n^{1/2} \log n) \text{ a.s. as } n \rightarrow \infty$$

$$(ii) \quad \sum_{i=1}^n I(g_1 \leq X_i \leq -X_{(s)} + a_n) = O(n^{1/2} \log n) \text{ a.s. as } n \rightarrow \infty$$

$$(iii) \quad \sum_{i=1}^n I(g_1 \leq X_i \leq g_1 + a_n) = O(n^{1/2} \log n) \text{ a.s. as } n \rightarrow \infty$$

$$(iv) \quad \sum_{i=1}^n I(g_2 - a_n \leq X_i \leq g_2) = O(n^{1/2} \log n) \text{ a.s. as } n \rightarrow \infty$$

Proof

We only consider (i), since (ii), (iii), and (iv) follow along the same lines.

$$\begin{aligned} & \sum_{i=1}^n I(-X_{(r)} - a_n \leq X_i \leq g_2) \\ &= nF_n(g_2) - nF_n(-X_{(r)} - a_n) \\ &= n(1 - \alpha) + O(n^{1/2}(\log_2 n)^{1/2}) - nF_n(-X_{(r)} - a_n) \end{aligned}$$

from the law of the iterated logarithm. But, by (iii) of Lemma 2

$$-X_{(r)} - a_n \in (g_2 - a_n^*, g_2 + a_n^*)$$

with $a_n^* \sim a_n \sim 2(\log n)/n^{1/2}$ as $n \rightarrow \infty$.

From (ii) of Lemma 2 we obtain

$$\begin{aligned}
 & nF_n(-X_{(r)} - a_n) \\
 &= nF_n(g_2) + nF(-X_{(r)} - a_n) - nF(g_2) + (n^{1/4} \log n) \\
 &= n(1 - \alpha) + O(n^{1/2}(\log_2 n)^{1/2}) + n[-X_{(r)} - a_n - g_2]f(g_2) + n[-X_{(r)} - a_n - g_2]^2 \\
 &\quad \cdot F''(\xi_n)/2
 \end{aligned}$$

by applying Taylor's theorem and the law of the iterated logarithm. Hence (i) follows from (A) and (C).

Define

$$(5) \quad G_n(x) = K^{-1} \sum_{i=r}^s \sum_{j=i+1}^s I((X_{(i)} + X_{(j)})/2 \leq x),$$

$$(6) \quad U_n(x) = \binom{n}{2}^{-1} \sum_{i=1}^n \sum_{j=i+1}^n I((X_i + X_j)/2 \leq x) I(g_1 \leq X_i \leq g_2) I(g_1 \leq X_j \leq g_2),$$

$$(7) \quad K = (s - r)(s - r + 1)/2, \text{ and}$$

$$(8) \quad \text{let } \{c_n: n = 1, 2, \dots\} \text{ be a sequence of positive constants such that } c_n \sim (\log n)/n^{1/2} \text{ and put } A_n = \sum_{i=1}^n I(X_i < g_1).$$

Theorem 1

For every $x \in [-c_n, +c_n]$ we have

$$KG_n(x) = \binom{n}{2} U_n(x) + (A_n - r)/(n - 2r) + R_n \text{ a.s. as } n \rightarrow \infty$$

where

$$R_n = O(n(\log_2 n)^{1/2} \log n) \text{ a.s. as } n \rightarrow \infty \text{ and independent of } x.$$

Proof

Note that

$$\begin{aligned}
 (9) \quad & \sum_{i=1}^n \sum_{j=i+1}^n I((X_{(i)} + X_{(j)})/2 \leq x) I(g_1 \leq X_{(i)} \leq g_2) I(g_1 \leq X_{(j)} \leq g_2) \\
 &= \binom{n}{2} U_n(x)
 \end{aligned}$$

From (9) it follows that (we use ideas analogous to those in De Wet and Venter (1974))

$$(10) \quad KG_n(x) = \binom{n}{2} U_n(x) + V_{1n} I(g_1 < X_{(r)}, X_{(s)} < g_2) + V_{2n} I(X_{(r)} < g_1, X_{(s)} < g_2) \\ + V_{3n} I(X_{(r)} < g_1, g_2 < X_{(s)}) + V_{4n} I(g_1 < X_{(r)}, g_2 < X_{(s)})$$

where

$$(11) \quad V_{1n} = -\sum_{i=1}^n \sum_{j=i+1}^n I((X_{(i)} + X_{(j)})/2 \leq x) I(g_1 \leq X_{(i)} < X_{(r)}) I(g_1 \leq X_{(j)} < X_{(r)}) \\ - \sum_{i=1}^n \sum_{j=i+1}^n I((X_{(i)} + X_{(j)})/2 \leq x) I(X_{(s)} < X_{(i)} \leq g_2) I(X_{(s)} < X_{(j)} \leq g_2) \\ - \sum_{i=1}^n \sum_{j=i+1}^n I((X_{(i)} + X_{(j)})/2 \leq x) I(g_1 \leq X_{(i)} < X_{(r)}) I(X_{(r)} \leq X_{(j)} \leq g_2) \\ - \sum_{i=1}^n \sum_{j=i+1}^n I((X_{(i)} + X_{(j)})/2 \leq x) I(X_{(r)} \leq X_{(i)} \leq X_{(s)}) I(X_{(s)} < X_{(j)} \leq g_2)$$

and where V_{2n} , V_{3n} and V_{4n} are given by similar expressions.

Let

$$(12) \quad B_n = \sum_{i=1}^n I(X_i > g_2).$$

On $\{g_1 < X_{(r)}, X_{(s)} < g_2\}$ we get:

$$(13) \quad V_{1n} \leq -\sum_{i < j} I((X_{(i)} + X_{(j)})/2 \leq x) I(g_1 \leq X_{(i)} < X_{(r)}) I(X_{(r)} \leq X_{(j)} \leq g_2) \\ = -(R-1-A_n)(n-B_n-r+1) + \sum_{i < j} I((X_{(i)} + X_{(j)})/2 > x) \\ I(g_1 \leq X_{(i)} < X_{(r)}) I(X_{(r)} \leq X_{(j)} \leq g_2) \\ \leq -(r-1-A_n)(n-B_n-r+1) + \sum_{i < j} I((X_{(i)} + X_{(j)})/2 > -c_n) \\ I(g_1 \leq X_{(i)} < X_{(r)}) I(X_{(r)} \leq X_{(j)} \leq g_2) \\ \leq -(r-1-A_n)(n-B_n-r+1) + \sum_{i < j} I(g_1 \leq X_{(i)} < X_{(r)}) \\ I(-2c_n - X_{(r)} \leq X_{(j)} \leq g_2) \\ = (A_n - r)(n - 2r) + O(n \log_2 n) + (r - 1 - A_n) O(n^{1/2} \log n)$$

$$= (A_n - r)(n - 2r) + O(n(\log_2 n)^{1/2} \log n)$$

by using (i) of Lemma 3 and the law of the iterated logarithm.

On $\{g_1 < X_{(r)}, X_{(s)} < g_2\}$ we have:

$$\begin{aligned}
 (14) \quad V_{1n} &\geq -(r-1-A_n)(r-2-A_n)/2 - (n-B_n-s)(n-B_n-s-1)/2 \\
 &\quad - (r-1-A_n)(n-B_n-r+1) \\
 &\quad + \sum_{i < j} I((X_{(i)} + X_{(j)})/2 > x) I(g_1 \leq X_{(i)} < X_{(r)}) I(X_{(r)} \leq X_{(j)} \leq g_2) \\
 &\quad - \sum_{i < j} I((X_{(i)} + X_{(j)})/2 \leq x) I(X_{(r)} \leq X_{(i)} \leq X_{(s)}) I(X_{(s)} < X_{(j)} \leq g_2) \\
 &\geq O(n \log_2 n) - (r-1-A_n)(n-B_n-r+1) \\
 &\quad - \sum_{i < j} I((X_{(i)} + X_{(j)})/2 \leq c_n) I(X_{(r)} \leq X_{(i)} \leq X_{(s)}) I(X_{(s)} < X_{(j)} \leq g_2) \\
 &\geq O(n \log_2 n) - (r-1-A_n)(n-B_n-r+1) \\
 &\quad - \sum_{i < j} I((X_{(i)} + X_{(j)})/2 \leq c_n) I(g_1 \leq X_{(i)} \leq X_{(s)}) I(X_{(s)} < X_{(j)} \leq g_2) \\
 &\geq O(n \log_2 n) - (r-1-A_n)(n-B_n-r+1) \\
 &\quad - \sum_{i < j} I(g_1 \leq X_{(i)} \leq -X_{(s)} + 2c_n) I(X_{(s)} < X_{(j)} \leq g_2) \\
 &= O(n \log_2 n) + (A_n - r)(n - 2r) - O(n^{1/2} \log n)(n - B_n - s) \\
 &= (A_n - r)(n - 2r) + O(n(\log_2 n)^{1/2} \log n)
 \end{aligned}$$

by making use of the law of the iterated logarithm and (ii) of Lemma 3. Consequently, from (13) and (14)

$$(15) \quad V_{1n} = (A_n - r)(n - 2r) + O(n(\log_2 n)^{1/2} \log n) \text{ a.s. on } \{g_1 < X_{(r)}, X_{(s)} < g_2\}$$

Similarly, upon use of Lemma 3, it may be shown that

$$(16) \quad V_{2n} = (A_n - r)(n - 2r) + O(n(\log_2 n)^{1/2} \log n) \text{ a.s. on } \{X_{(r)} < g_1, X_{(s)} < g_2\}$$

$$(17) \quad V_{3n} = (A_n - r)(n - 2r) + O(n(\log_2 n)^{1/2} \log n) \text{ a.s. on } \{X_{(r)} < g_1, g_2 < X_{(s)}\}$$

$$(18) \quad V_{4n} = (A_n - r)(n - 2r) + O(n(\log_2 n)^{1/2} \log n) \text{ a.s. on } \{g_1 < X_{(r)}, g_2 < X_{(s)}\}$$

The proof of the theorem follows from (15), (16), (17) and (18).

Remark

If $x = 0$ the above result can be improved (by repeating the arguments of Theorem 1) to yield a lower order remainder term (compare De Wet and Venter (1974)).

$$(19) \quad KG_n(0) = \left(\frac{n}{2}\right) U_n(0) + (n - 2r)(A_n - r) + O(n \log_2 n) \text{ a.s. as } n \rightarrow \infty.$$

If we define the trimmed Wilcoxon one-sample statistic by

$$(20) \quad \tilde{G}_n(0) = \tilde{K}^{-1} \sum_{i=r}^s \sum_{j=i}^s I((X_{(i)} + X_{(j)})/2 \leq 0)$$

where

$$\tilde{K} = (s - r + 1)(s - r + 2)/2$$

then, using the fact that $\tilde{G}_n(0) = G_n(0) + O(n^{-1})$, we obtain

Corollary 1

$$(21) \quad \tilde{G}_n(0) - \alpha = n^{-1} \sum_{i=1}^n Y_i + O(n^{-1} \log n) \text{ a.s. as } n \rightarrow \infty$$

where

$$Y_i = 2(1 - 2\alpha)^{-2}$$

$$\{[\alpha - \frac{1}{2}]I(X_i > g_2) - [F(X_i) - \frac{1}{2}]I(g_1 \leq X_i \leq g_2) - [\alpha - \frac{1}{2}]I(X_i < g_1)\}$$

Proof

Since $U_n(0)$ is a one-sample U-statistic, it follows by applying Corollary 4.2 of Geertsema (1970), that

$$(22) \quad U_n(0) = n^{-1} \sum_{j=1}^n h(X_j) + o(n^{-1} \log n) \text{ a.s. as } n \rightarrow \infty$$

where

$$(23) \quad h(x) = 2E \{I(X_1 + X_2 \leq 0)I(g_1 \leq X_1 \leq g_2)I(g_1 \leq X_2 \leq g_2) | X_1 = x\} - U(0) \\ = 2I(g_1 \leq x \leq g_2)[F(-x) - \alpha] - (1 - 2\alpha)^2/2.$$

The proof of (21) then follows from (19), (20), (22) and (23).

Let $W_{(1)} \leq W_{(2)} \leq \dots \leq W_{(K)}$ be the ordered averages $\{X_{(i)} + X_{(j)}\} / 2$ for $r \leq i < j \leq s$. In the following theorem it is proved that Geertsema's result (1970) can be extended to apply also to the ordered averages of pairs of observations that are being considered.

Theorem 2

If $\{k(n)\}$ is any sequence of integers satisfying $k(n)/K = \frac{1}{2} + o(n^{-1/2} \log n)$ as $n \rightarrow \infty$ then, with probability one,

$$(24) \quad W_{(k(n))} = \{k(n)/K - G_n(0)\} / 2(1 - 2\alpha)^{-2} \int_{g_1}^{g_2} f^2(x) dx + o(n^{-1/2} \log n) \text{ as } n \rightarrow \infty.$$

Proof

Let $D_n(x) = [G_n(x) - G_n(0)] - [(1 - 2\alpha)^{-2}U(x) - (1 - 2\alpha)^{-2}U(0)]$. $\{c_n\}$ is a sequence of constants with $c_n \sim (\log n)/n^{1/2}$ as $n \rightarrow \infty$. $I_n = [-c_n, +c_n]$, $E_n = \sup \{|D_n(x)| : x \in I_n\}$. First prove that

$$(25) \quad E_n = O(n^{-1/2} \log n) \text{ a.s. as } n \rightarrow \infty.$$

Using Theorem 1 and the fact that $1 - K/\binom{n}{2}(1 - 2\alpha)^2 = O(n^{-1})$, it suffices to prove

$$(26) \quad H_n = O(n^{-1/2} \log n) \text{ a.s. as } n \rightarrow \infty,$$

where

$$(27) \quad H_n = \sup \{|B_n(x)| : x \in I_n\}, B_n(x) = [U_n(x) - U_n(0)] - [U(x) - U(0)]$$

Since $U_n(x)$ is a one-sample U-statistic the proof of (26) now follows in a similar way as (4.7) of Geertsema (1970). The second part of the proof consists in proving that

$$(28) \quad \text{With probability one, } W_{(k(n))} \text{ is in } I_n, \text{ for all sufficiently large } n.$$

From (20), (21) and the law of the iterated logarithm it follows

Applying Taylor's theorem we get

$$(30) \quad U(-c_n) = U(0) - c_n U'(0) + c_n^2 U''(\xi_n)/2 \text{ where } -c_n \leq \xi_n \leq 0 \\ = U(0) - c_n U'(0) + o(c_n)$$

because U'' is bounded in a neighbourhood of 0. Also, from (25)

$$(31) \quad G_n(-c_n) = G_n(0) + (1 - 2\alpha)^{-2} U(-c_n) - (1 - 2\alpha)^{-2} U(0) + O(n^{-\frac{1}{2}} \log n) \\ \text{a.s. as } n \rightarrow \infty$$

From the definition of $\{k(n)\}$, (29), (30), (31) and the fact that

$$(32) \quad U'(0) = 2 \int_{g_1}^{g_2} f^2(x) dx > 0$$

we have

$$(33) \quad W_{(k(n))} \geq -c_n \text{ iff} \\ G_n(-c_n) \leq k(n)/K = \frac{1}{2} + o(c_n) \text{ iff} \\ -c_n U'(0)(1 + o(1)) \leq 0 \text{ a.s.}$$

A similar argument gives

$$(34) \quad W_{(k(n))} \leq +c_n \text{ a.s. for all sufficiently large } n.$$

The proof of (28) follows from (33) and (34).

The proof of (24) can now be completed. By (28), with probability one, there is an n_2 such that $W_{(k(n))}$ is in I_n for $n > n_2$. Also n_2 can be chosen such that for $n > n_2$ $U''(x)$ is defined and bounded on I_n . Let p_3 be such that

$$\frac{1}{2} |U''(x)| \leq p_3 \text{ for } x \text{ in } I_n \text{ and } n > n_2.$$

Now $G_n(W_{(k(n))}) = k(n)/K$ so that, by definition of $D_n(x)$,

$$(35) \quad k(n)/K = G_n(0) + (1 - 2\alpha)^{-2} U(W_{(k(n))}) - (1 - 2\alpha)^{-2} U(0) + \theta_n E_n \text{ with } |\theta_n| \leq 1.$$

$$U(W_{(k(n))}) = U(0) + W_{(k(n))} U'(0) + W_{(k(n))}^2 (\xi_n)/2$$

with ξ_n between 0 and $W_{(k(n))}$

$$= U(0) + W_{(k(n))} U'(0) + p_3 \phi_n c_n^2 \text{ with } |\phi_n| \leq 1.$$

Substitute in (35). Then one obtains (from (32))

$$k(n)/K = G_n(0) + W_{(k(n))} 2(1 - 2\alpha)^{-2} \int_{g_1}^{g_2} f^2(x) dx + O(n^{-\frac{3}{2}} \log n) \text{ a.s.}$$

This completes the proof.

Remarks

- (i) The method of proof of (28) given here differs from the method of Bahadur (1966, Lemma 2) and Geertsema (1970, (4.8)).
- (ii) By repeating the arguments of Theorem 2 it follows that (24) continues to hold if $W_{(k(n))}$, K and $G_n(0)$ are replaced by $\tilde{W}_{(k(n))}$, $\tilde{K}$ and $\tilde{G}_n(0)$, where $\tilde{W}_{(1)} \leq \tilde{W}_{(2)} \leq \dots \leq \tilde{W}_{(\tilde{K})}$ are the ordered averages $\{X_{(i)} + X_{(j)}\}/2$ for $r \leq i \leq j \leq s$.

Denote by M_n the trimmed Hodges-Lehmann estimator, viz. the median of $\tilde{W}_{(1)}, \dots, \tilde{W}_{(\tilde{K})}$.

Corollary 2

$$(36) \quad M_n = n^{-1} \sum_{i=1}^n Y_i^* + O(n^{-\frac{3}{2}} \log n) \text{ a.s. as } n \rightarrow \infty,$$

where

$$Y_i^* = \{[F(X_i) - \frac{1}{2}]\}$$

$$\cdot I(g_1 \leq X_i \leq g_2) + [\alpha - \frac{1}{2}] I(X_i < g_1) - [\alpha - \frac{1}{2}] I(X_i > g_2) \} / \int_{g_1}^{g_2} f^2(x) dx.$$

Proof

From (21) and (ii) the proof follows directly.

Asymptotic efficiencies

. From (36) it follows that $n^{1/2}M_n$ is asymptotically $N(0, \sigma^2(\alpha))$ distributed with

$$(37) \quad \sigma^2(\alpha) = (1 - 2\alpha)^2(1 + 4\alpha) / \{12(\int_{g_1}^{g_2} f^2(x)dx)^2\}.$$

Let

$$(38) \quad e(\alpha) = \sigma^2(0)/\sigma^2(\alpha).$$

Values of $e(\alpha)$ were tabulated by Hettmansperger (1968) for three different distributions and for various values of α . Figure 1.1 shows the values of $e(\alpha)$ for various values of α and γ for the contaminated normal distribution with density $(1 - \gamma)\phi(x) + \gamma\phi(x/3)/3$, where $\phi(x)$ is the standard normal density function.

From De Wet and Venter (1974) it follows that the α -trimmed mean is asymptotically $N(0, \sigma_M^2(\alpha))$ distributed with

$$(39) \quad \sigma_M^2(\alpha) = (1 - 2\alpha)^{-2} [\int_{g_1}^{g_2} x^2 dF(x) + 2\alpha g_1^2].$$

Let

$$(40) \quad e^*(\alpha) = \sigma_M^2(\alpha)/\sigma^2(\alpha).$$

TABLE 1.1

Values of $e^*(\alpha) = \sigma_M^2(\alpha)/\sigma^2(\alpha)$

α	0,05	0,10	0,15	0,20	0,25	0,30	0,35	0,40	0,45	0,49
Double exponential	1,251	1,153	1,096	1,059	1,036	1,020	1,010	1,004	1,001	1,000
Cauchy	2,734	1,577	1,236	1,098	1,037	1,011	1,002	1,000	1,000	1,000
Normal	0,968	0,978	0,985	0,990	0,994	0,997	0,999	1,000	1,000	1,000

Table 1.1 gives values of $e^*(\alpha)$ for various values of α and for the following populations

- (i) Double exponential with density $\frac{1}{2}\exp(-|x|)$,

- (ii) Cauchy with density $[\pi(1 + x^2)]^{-1}$, and
- (iii) Standard normal

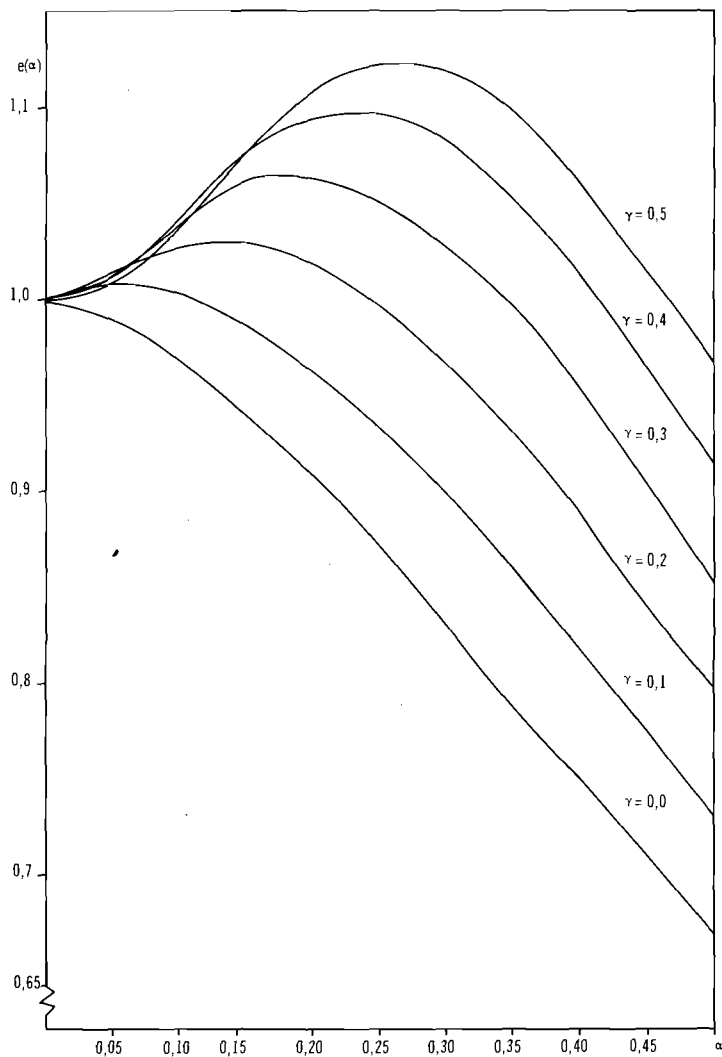


FIGURE 1.1

2. A bounded length sequential confidence interval

In this section we consider the problem of providing a fixed-width sequential confidence interval for the median θ .

(41) Let N be the first integer $n \geq n_0$ for which $\tilde{W}_{(a(n))} - \tilde{W}_{(b(n))} \leq 2d$ and choose as resulting confidence interval $I_N = [\tilde{W}_{(b(N))}, \tilde{W}_{(a(N))}]$. n_0 is some positive integer, $\{a(n)\}$ and $\{b(n)\}$ are sequences of positive integers satisfying:

(42) $a(n) \sim \tilde{K}/2 + \tilde{K}(1 + 4\alpha)^{1/2} \delta_\beta / (1 - 2\alpha)(3n)^{1/2}$,
 $b(n) \sim \tilde{K}/2 - \tilde{K}(1 + 4\alpha)^{1/2} \delta_\beta / (1 - 2\alpha)(3n)^{1/2}$ as $n \rightarrow \infty$,
 and $\Phi(\delta_\beta) = 1 - \beta$ where Φ is the standard normal cdf.

Theorem 3

- (i) $P(N < \infty) = 1$, $N(=N(d))$ is a function of d which is non-decreasing as d decreases, $\lim_{d \rightarrow 0} N = \infty$ a.s. and $\lim_{d \rightarrow 0} E(N) = \infty$.
- (ii) $\lim_{d \rightarrow 0} Nd^2 = \delta_\beta^2 / A^2$ a.s.
- (iii) $\lim_{d \rightarrow 0} P\{\theta \in I_N\} = 1 - 2\beta$,

where $A = 12^{1/2} \int_1^{\delta^2} f^2(x) dx / (1 - 2\alpha)(1 + 4\alpha)^{1/2}$.

Remarks

- (a) The procedure of Geertsema (1970, p. 1020) is obtained by taking $\alpha = 0$ in (41).
 - (b) If the set $\{Nd^2\}_{d>0}$ is uniformly integrable it follows from (ii) of Theorem 3 that
- (43) $\lim_{d \rightarrow 0} E(N)d^2 = \delta_\beta^2 / A^2$.

Monte Carlo studies

Important questions that arise are: how well do the above asymptotic results describe the behaviour of the procedure as to coverage probability and expected sample size?

The above questions were investigated by means of limited Monte Carlo studies for

the contaminated normal population with density function $0,5\phi(x) + 0,5\phi(x/3)/3$. A sequence of 5 values of d was chosen in such a way that d^{-2} increases by a constant amount. Also β was chosen 0,025 (so that $1 - 2\beta = 0,95$). The procedure was required to take at least ten observations because $P(\tilde{W}_{(b(n))} \leq \theta \leq \tilde{W}_{(a(n))}) \geq 0,95$ for the first time when $n = 10$.

Table 2.1 gives, for each fixed d and α , the estimated coverage probability ($\hat{P}$) and the standard error (SE) of the $\hat{P}$ values. Table 2.2 gives the estimated expected sample size ($\hat{EN}$) and the standard error of the $\hat{EN}$ -values between parentheses. The asymptotic expected sample size (AS), computed from (43), is also given in table 2.2. For each fixed d and α , the Monte Carlo results are based on 500 repetitions.

TABLE 2.1

2d	$\alpha = 0,00$		$\alpha = 0,005$		$\alpha = 0,15$	
	$\hat{P}$	SE	$\hat{P}$	SE	$\hat{P}$	SE
1,794	0,962	0,009	0,960	0,009	0,954	0,009
1,268	0,952	0,010	0,962	0,009	0,960	0,009
1,036	0,954	0,009	0,950	0,009	0,958	0,009
0,897	0,946	0,010	0,958	0,007	0,962	0,009
0,802	0,944	0,018	0,956	0,008	0,964	0,008

TABLE 2.2

2d	$\alpha = 0,0$		$\alpha = 0,05$		$\alpha = 0,15$	
	$\hat{EN}$	AS	$\hat{EN}$	AS	$\hat{EN}$	AS
1,794	25,1 (0,35)	26,1	23,1 (0,32)	26,0	25,4 (0,40)	25,0
1,268	40,4 (0,54)	42,3	38,1 (0,51)	42,0	40,3 (0,56)	40,0
1,036	55,8 (0,69)	58,3	53,6 (0,75)	58,0	55,3 (0,71)	54,9
0,897	71,7 (0,79)	74,3	68,0 (0,88)	73,9	68,9 (0,81)	69,8
0,802	89,2 (0,84)	90,7	84,0 (0,89)	90,1	84,2 (0,86)	84,9

The following may be concluded:

- (i) From Table 2.1 it follows that the actual coverage probability is quite close to the asymptotic coverage probability and in most cases the actual coverage probability for the two procedures with $\alpha = 0,05$ and $\alpha = 0,15$ is higher than that of the procedure with $\alpha = 0$. Similar results were obtained by Starr (1966) and Geertsema (1970).
- (ii) From Table 2.2 it follows that the EN-values of the procedure with $\alpha = 0,05$ are smaller than the corresponding values of the procedures with $\alpha = 0$ and $\alpha = 0,15$. This fact does not follow from the asymptotic theory discussed above. Also, the Monte Carlo results seem to indicate that the expression for the asymptotic expected sample size (AS-value) is in fact an upper bound for the actual expected sample size.

3. A sequential selection procedure

It is also possible to define a sequential selection procedure for the problem of selecting the population with the largest location parameter, based on the trimmed Hodges-Lehmann estimator (see Corollary 2), similar to the procedure of Robbins, Sobel and Starr (1968).

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